

Machine Learning: Important Topics for Freshers

A structured beginner-friendly roadmap covering the concepts, tools, mathematics, and practical skills every ML fresher should know.

1. ML Basics

Understand AI vs ML vs Deep Learning, supervised learning, unsupervised learning, and reinforcement learning. Be clear about features, labels, training, testing, prediction, and model.

2. Data Preprocessing

Learn how to handle missing values, duplicate records, and outliers. Understand categorical encoding, normalization vs standardization, and train/validation/test split.

3. Regression Algorithms

Study Linear Regression, Multiple Linear Regression, Polynomial Regression, Ridge Regression, and Lasso Regression. Know evaluation metrics such as MAE, MSE, RMSE, and R-squared.

4. Classification Algorithms

Learn Logistic Regression, K-Nearest Neighbors (KNN), Decision Tree, Random Forest, Naive Bayes, and Support Vector Machine (SVM). Understand confusion matrix, accuracy, precision, recall, F1-score, and ROC-AUC.

5. Unsupervised Learning

Understand K-Means Clustering, Hierarchical Clustering, the basic idea of DBSCAN, and PCA for dimensionality reduction.

6. Core ML Concepts

Be comfortable with overfitting vs underfitting, bias-variance tradeoff, regularization, cross-validation, hyperparameter tuning, data leakage, and class imbalance.

7. Feature Engineering

Learn feature selection, feature creation, scaling, encoding, and the basic interpretation of feature importance.

8. Essential Tools

Build working knowledge of Python, NumPy, Pandas, Matplotlib, and scikit-learn. Also learn the basics of Jupyter Notebook and Git/GitHub.

9. Mathematics for ML

Cover mean, median, variance, standard deviation, probability basics, common distributions, correlation vs causation, vectors and matrices, and the intuition behind derivatives and gradient descent.

10. Projects and Interview Skills

Practice the complete workflow: EDA, preprocessing, model training, evaluation, improvement, and explanation. A fresher should be able to clearly explain at least 2-3 end-to-end ML projects.

Recommended Learning Order

- 1 Python
- 2 NumPy and Pandas
- 3 Statistics
- 4 Data Preprocessing and EDA
- 5 Regression
- 6 Classification
- 7 Evaluation Metrics
- 8 Trees and Ensemble Methods
- 9 Clustering and PCA
- 10 Feature Engineering
- 11 Model Tuning
- 12 End-to-End Projects

High-Priority Interview Topics

- Overfitting and underfitting
- Train-test split and cross-validation
- Precision vs recall
- Confusion matrix
- Linear Regression vs Logistic Regression
- Decision Tree vs Random Forest
- Feature scaling
- Categorical encoding
- Regularization
- Data leakage

Practical advice: Do not only memorize algorithm definitions. For every important algorithm, learn when to use it, how it works at a high level, its main hyperparameters, its advantages and limitations, and how to evaluate its performance.